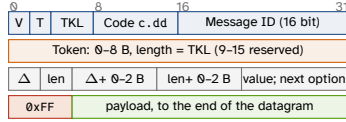


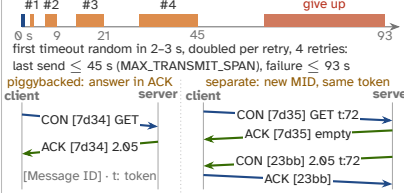
In one line: CoAP (RFC 7252, 2014) is **REST over UDP**: methods and HTTP-like codes in a **4-byte binary header**, per-message reliability (**CON** + ACK, back-off), push by **Observe**, big bodies **block-wise**, security by **DTLS** (per hop) or **OSCORE** (end to end). **LwM2M** (OMA SpecWorks) is **device management on CoAP**: a numbered **Object** / **Instance** / **Resource** tree on the device (/3/0/3 = firmware version) plus bootstrap, registration, management and reporting.

CoAP over UDP — the message

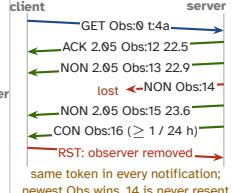


Ver = 1 · T: 0 CON · 1 NON · 2 ACK · 3 RST
Code = class (3 bit) · detail (5 bit); 0:00 = Empty options ascend by number, each stores the delta; nibble 13 → +1 B (value -13) · 14 → +2 B (-269) · 15 is reserved: the byte 0xFF ends the options

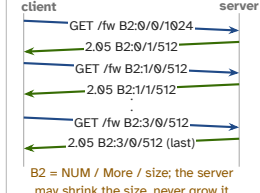
CON retransmission — default timers



Observe (RFC 7641)



Block-wise GET (RFC 7959)



Messages, timers, codes

- CON** is resent until an ACK or RST with its Message ID arrives; **NON** is never ACKed. Empty CON (0:00) → RST = "CoAP ping". **NSTART 1**: one open interaction per server.
- Message ID** pairs CON with ACK and deduplicates: not reused towards a peer within EXCHANGE_LIFETIME 247 s ($2^{16} / 247 \text{ s} \approx 265 \text{ CON/s per peer}$); a duplicate gets the same ACK and is processed once. **Token** (0-8 B, the client's choice) matches responses and notifications to the request: ≥ 32 random bits on the Internet, or a per-session counter (RFC 9175).
- Multicast**: NON only; replies spread over the 5 s leisure. **Size**: unknown MTU → assume 1280 B: message $\leq 1152 \text{ B}$, payload $\leq 1024 \text{ B}$ (802.15.4 frame: 127 B).
- Timers**: ACK_TIMEOUT 2 s, ACK_RANDOM_FACTOR 1.5, MAX_RETRANSMIT 4, PROBING_RATE 1 B/s $\Rightarrow$ MAX_LATENCY 100 s, PROCESSING_DELAY 2 s, MAX_RTT 202 s, NON_LIFETIME 145 s.

0.xx 0:01 GET · 0:02 POST · 0:03 PUT · 0:04 DELETE · 0:05 FETCH · 0:06 PATCH · 0:07 IPATCH
2.xx 2:01 Created · 2:02 Deleted · 2:03 Valid (ETag) · 2:04 Changed · 2:05 Content · 2:31 Continue
4.xx 4:00 Bad Request · 4:01 Unauthorized · 4:02 Bad Option · 4:03 Forbidden · 4:04 Not Found ·
4:05 Method Not Allowed · 4:06 Not Acceptable · 4:08 Request Entity Incomplete · 4:09 Conflict · 4:12 Precondition Failed · 4:13 Request Entity Too Large · 4:15 Unsupported Content-Format · 4:22 Unprocessable Entity · 4:29 Too Many Requests
5.xx 5:00 Internal Server Error · 5:01 Not Implemented · 5:02 Bad Gateway · 5:03 Service Unavailable · 5:04 Gateway Timeout · 5:05 Proxying Not Supported · 5:08 Hop Limit Reached
7.xx TCP / WS only: 7:01 CSM · 7:02 Ping · 7:03 Pong · 7:04 Release · 7:05 Abort

Options and formats

Odd number = **critical**: unknown in a CON request → 4:02 Bad Option (even = elective, ignored). Bit 1 set = **unsafe** for a proxy that does not know it; (n & 0x1e) == 0x1c = not part of the cache key. Max-Age defaults to 60 s.

1 If-Match	9 OSCORE	19 Q-Block1	35 Proxy-Uri
3 Uri-Host	11 Uri-Path	20 Location-Query	39 Proxy-Scheme
4 ETag	12 Content-Format	21 EDHOC	60 Size1
5 If-None-Match	14 Max-Age	23 Block2	252 Echo
6 Observe	15 Uri-Query	27 Block1	258 No-Response
7 Uri-Port	16 Hop-Limit	28 Size2	292 Request-Tag
8 Location-Path	17 Accept	31 Q-Block2	

Content-Format: 0 text · 40 link-format · 50 JSON · 60 CBOR · 61 CWT · 110 / 112 SenML JSON / CBOR · 1000/01 OSCORE · 11542-4 LwM2M TLV / JSON / CBOR.

Observe · block-wise · discovery · TCP

Observe (6): GET + Observe 0 registers endpoint + token; notifications are 2.05 on that token with a 24-bit counter; cancel by GET + Observe 1, RST, or forgetting.
Eventually consistent: states may be skipped; V2 is newer if $0 < (V2 - V1) \bmod 2^{24} < 2^{23}$ or it came $> 128 \text{ s}$ later. RST or a timed-out CON drops the observer.

Block-wise: Block2 (23) = response body, Block1 (27) = request body; NUM | M | SZX in 1-3 B (NUM 4 / 12 / 20 bit), size $2^{\text{SZX}+4} = 16 \dots 1024 \text{ B}$; SZX 7 reserved on UDP, **BERT** on TCP (n $\times$ 1024 B). Atomic upload: 2.31 Continue per block, gap → 4.08, too large → 4.13 + Size1. **Request-Tag** keeps two bodies apart; **Q-Block1/2** (RFC 9177): NON bursts, all gaps re-asked at once (DOTS, lossy links).

Discovery: GET /.well-known/core → link format, e.g.
</sensors/temp>;if="sensor": rt type (noun), if interface (verbs), sz max size, ct format; ?rt=temp* filters by prefix. **Resource Directory** (RFC 9176): POST <rd?ep=...<=... + links, default lifetime 90 000 s; found via ?rt=core.rd*.

TCP · TLS · WebSockets (RFC 8323): coap+tcp 5683, coaps+tcp 5684 (ALPN coap), coap(s)+ws 80 / 443 at /.well-known/coap; Len + TKL byte, no Type, no MID; 7:01 CSM states the max message size. RFC 8974 tokens ($\leq 65 804 \text{ B}$) hold a stateless proxy's state. **vs MQTT** (TCP 1883 / 8883, $\geq 2 \text{ B}$ header, QoS 0-2): pub / sub via a broker on the device's own TCP; CoAP has no broker — the device is a server.

Traps

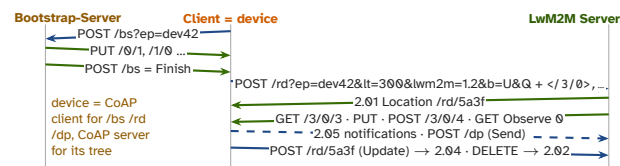
- ! Matching responses by Message ID — a separate one has a new MID.
- ! Observe as an event log — it skips states; Send or hqmax keep history.
- ! Idle UDP to a device behind NAT — the mapping may expire after 2 min (RFC 4787 floor, 5 min recommended), silently; DTLS 1.2 without CID then re-handshakes.
- ! CoAP over TCP has no CON / NON / RST — cancel an Observe with GET Observe 1.

Security — the hop or the message

	DTLS 1.2 RFC 6347 · 1.3 RFC 9147	OSCORE RFC 8613 (2019)
protects proxy	one hop, the whole datagram ends it, reads everything	end to end: payload + class-E options forwards; instance Uri-Host, Proxy-Uri, outer Observe
keys	PSK, raw public key or X.509 (else NoSec)	Master Secret + IDs → HKDF-SHA-256, or EDHOC
must have new address	PSK and ECDHE-ECDSA, AES-128-CCM-8 1.2 re-handshakes unless CID (RFC 9146); 1.3 has CID	AES-CCM-16-64-128 no effect; replay window of 32

DTLS 1.3: short header, ACK record, HelloRetryRequest cookie checks the address, timer 1 s doubling. **OSCORE**'s outer POST → 2.04 (FETCH) → 2.05 when observing) hides the method. **EDHOC** (RFC 9528, 2024): 3 messages of ephemeral DH over COSE → OSCORE keys; option 21 (RFC 9668) adds message_3 to the first OSCORE request. **Echo** (252) proves freshness and address; until then $\leq 3 \times$ amplification.

LwM2M — roles, model, operations

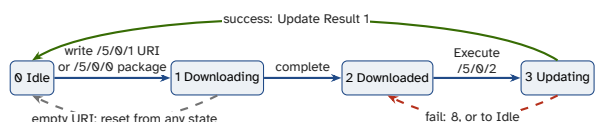


Path /Object/Instance/Resource/Resource-Instance (4th level since 1.1), 16-bit IDs, 65535 reserved: /3303/0/5700 = Temperature, instance 0, Sensor Value. **Objects**: 0 Security · 1 Server (1 Lifetime = 1 t) · 2 Access Control · 3 Device (0 Manufacturer, 3 FW Version, 4 Reboot = E) · 4 Connectivity Monitoring · 5 Firmware Update · 6 Location · 9 Software Management · 21 OSCORE · 25 Gateway.

operation	on CoAP
Read · Discover · Write	GET · GET Accept 40 · PUT replaces, POST updates part
Write-Attributes	PUT /3303/0/5700?pmIn=10&pmax=60>=30
Execute · Create · Delete	POST /3/0/4 · POST /3303 · DELETE /3303/1
Composite (1.1)	Read = FETCH (a GET with a body), Write = IPATCH; body = paths (+ values)

Attributes (Write-Attributes; in the Observe since 1.2): pmin / pmax = min / max seconds between notifications; gt / lt / st = threshold / step, held back by pmin; epmin / epmax = evaluation periods; 1.2: edge, con (CON notifications), hqmax (history kept offline). **Bindings** b=: U UDP (default) · T TCP · S SMS · N Non-IP (3GPP CioT, LoRaWAN) · M MQTT · H HTTP (M, H since 1.2). **Queue mode** (Q): the server holds requests until the sleeping device speaks; it should stay up $\geq \text{MAX_TRANSMIT_WAIT}$ (93 s) after its last message. **Security Mode** /0/x/2: 0 PSK · 1 RPK · 2 Certificate · 3 NoSec · 4 Certificate + EST; DTLS 1.2 mandatory, (D)TLS 1.3 + CID optional (1.2). **Formats**: MUST plain text, opaque, CoRE link; SHOULD SenML or LwM2M CBOR.

Firmware update — Object 5



Update Result: 2 no flash · 3 no RAM · 4 link lost · 5 integrity (→ Idle) · 6 package type · 7 URI · 9 protocol · 10 cancelled · 11 deferred.

Versions

7252	7641	7959	8132	8323	8613	8974	9147	9528
CoAP	Observe	Block	FETCH	TCP	OSCORE	tokens	DTLS 1.3	EDHOC
2014	LwM2M: 1.0	1.1	1.2	1.2.1	1.2.2	2026		

LwM2M 1.1 (Jul 2018): TCP / TLS, OSCORE, LPWAN, Resource Instances, composite, SenML, Send. **1.2** (Nov 2020): MQTT + HTTP, Bootstrap-Pack, Gateway, LwM2M CBOR, edge / con / hqmax, (D)TLS 1.3. Latest approved **1.2.2** (13 Jun 2024); no 2.0 in OMA's release directory on 9 Oct 2026. **Leshan** (Java): 1.x = LwM2M 1.0, 2.0.0-M18 (Dec 2025) = 1.1; **Wakaama** (C): repository archived 15 Sep 2026.